User's
Manual

YS1000 Series

YS1350 Manual Setter for SV Setting
YS1360 Manual Setter for MV Setting
User's Manual

Functional
Enhancement

IM 01B08E02-02EN

YOKOGAWA

Yokogawa Electric Corporation

IM 01B08E02-02EN
6th Edition

Introduction

Thank you for purchasing the YS1000 series single-loop controller (hereinafter referred to as "YS1000").
This manual describes how to use YS1350 and YS1360 functions other than communication function. Please read through this user's manual carefully before using the product.
Note that the manuals for the YS1350/YS1360 comprise the following five documents:

● Printed manual

Manual Name	Manual Number	Description
YS1350/YS1360 Operation Guide	IM 01B08E02-01EN	This manual describes the basic operation method.

● Electronic manuals

Manual Name	Manual Number	Description
YS1350/YS1360 Operation Guide	IM 01B08E02-01EN	This is identical to the printed manual.
YS1350/YS1360 User's Manual	IM 01B08E02-02EN	This manual. It describes the usage of all functions except the communication functions.
YS1000 Series Communication Interface User's Manual	IM 01B08J02-01EN	This manual describes how to use YS1000 in Ethernet, serial, and DCS-LCS communications. For communication wiring, see the Operation Guide.
YSS1000 Setting Software/YS1700 Programmable Function User's Manual	IM 01B08K02-02EN	This manual describes how to use YSS1000 and YS1700's programmable function and peer-to-peer communication function.
YS1000 Series Replacement Manual	IM 01B08H02-01EN	This manual describes the compatibility of installation and wiring with YS100, YS80, EBS, I, EK, HOMAC, and 100 line.

User's manuals for YS1000 are available on the following web site: www.yokogawa.com/ns/ys/im/
You need Adobe Reader 7.0 or later (but the latest version is recommended) installed on the computer in order to open and read the manuals.

Notice

- The contents of this manual are subject to change without notice as a result of continuing improvements to the instrument's performance and functions.
- Every effort has been made to ensure accuracy in the preparation of this manual. Should any errors or omissions come to your attention, however, please inform YOKOGAWA Electric's sales office or sales representative.
- Under no circumstances may the contents of this manual, in part or in whole, be transcribed or copied without our permission.

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6th Edition : May 2025 (YK)

How to Use This Manual

Usage

First read through the Operation Guide to understand the basic operations and then read this manual. For the communication functions and replacements, see the respective manuals.
This User's Manual is organized into Chapters 1 to 6 as shown below:

Chapter	Title and Description
1	Input/output and Auxiliary Function Describes the input/output function, operation and setting items.
2	Display Function/Security Function Provides the LCD display functions, adjustments, and setting items.
3	Adjustment of Direct Inputs (Temperature, Resistance, and Frequency) Describes the settings and adjustments for the direct input converter built into the YS1350/1360.
4	Power Failure Recovery Processing Describes operations performed after momentary power interruption and power failures.
5	Maintenance Describes ordinary inspections, indicating accuracy inspections, and part replacement cycles.
6	Specifications For specifications, see the YS1350 Manual Setter for SV Setting YS1360 Manual Setter for MV Setting Operation Guide (IM 01B08E02-01EN).

Symbols Used in This Manual

⚠

This symbol is used on the instrument. It indicates the possibility of injury to the user or damage to the instrument, and signifies that the user must refer to the user's manual for special instructions. The same symbol is used in the user's manual on pages that the user needs to refer to, together with the term "WARNING" or "CAUTION."

WARNING

Calls attention to actions or conditions that could cause serious or fatal injury to the user, and indicates precautions that should be taken to prevent such occurrences.

CAUTION

Calls attention to actions or conditions that could cause injury to the user or damage to the instrument or property and indicates precautions that should be taken to prevent such occurrences.

Note

Identifies important information required to operate the instrument.

▶

Indicates related operations or explanations for the user's reference.

[ ]

Indicates a character string displayed on the display.

Setting Display

Indicates a setting display and describes the keystrokes required to display the relevant setting display.

Setting Details

Provides the descriptions of settings.

Description

Describes restrictions, etc. regarding a relevant operation.

QR Code

The product has a QR Code pasted for efficient plant maintenance work and asset information management. It enables confirming the specifications of purchased products and user's manuals.
For more details, please refer to the following URL.
https://www.yokogawa.com/qr-code
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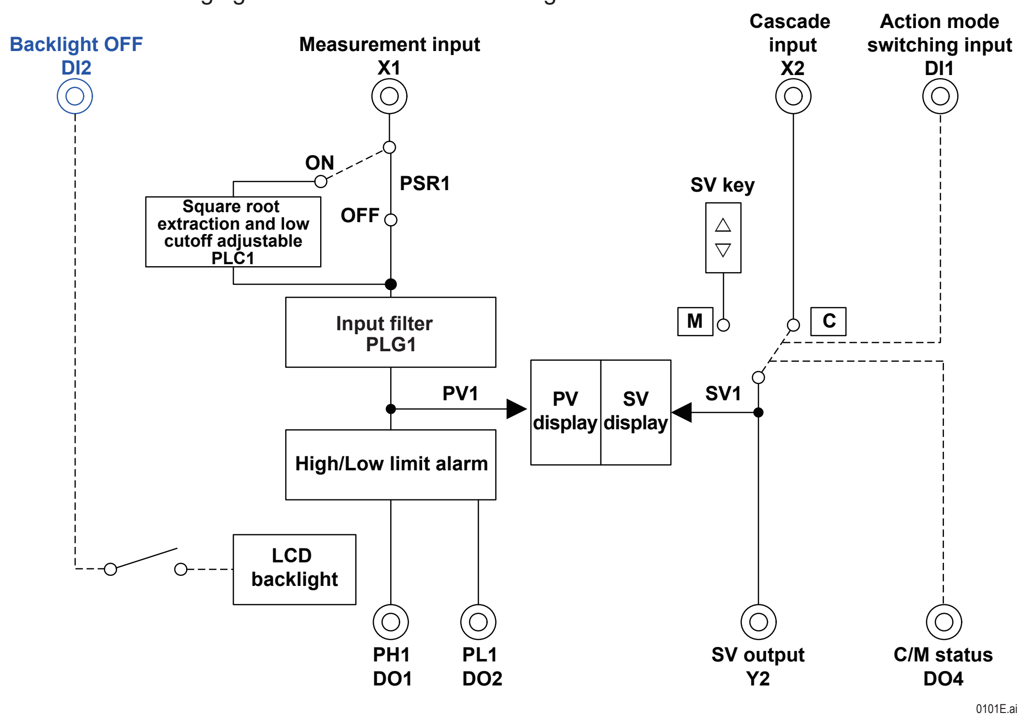
Contents

1.1 Function Block Diagram

1.1.1 YS1350

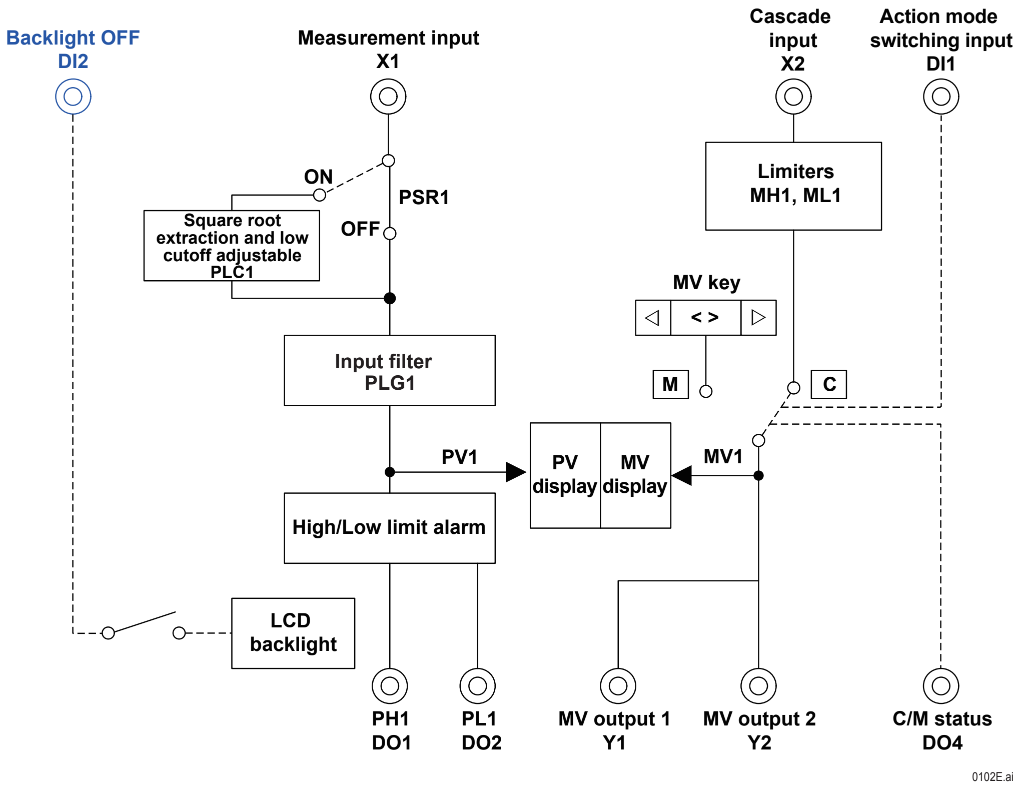
The following figure shows the function configuration.



Function Block Diagram of YS1350

1.1.2 YS1360

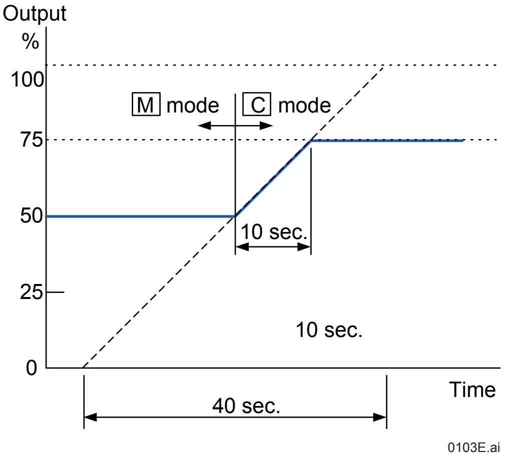
The following figure shows the function configuration.



Function Block Diagram of YS1360

■ Ramp Follow-up (YS1350/YS1360)

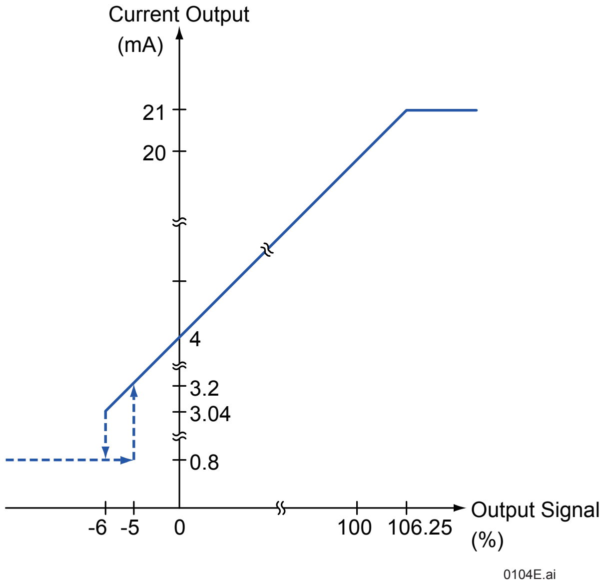
The YS1350 manual setter for SV setting and YS1360 manual setter for MV setting switch the ramp characteristic bumplessly when switching from mode M to mode C.
The switching rate is 40 seconds/full scale.
For example, if 50% output in mode M is switched to 75% output in mode C as shown in the figure below, the output reaches the 75% level in 10 seconds following a gradient of being full scale in 40 seconds.



Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
CMOD1	C-mode 1	-: None, CAS: Analog cascade setting mode, CMP: Computer cascade setting mode	Tuning Display > Engineering Display > [**CONFIG2**] (Configuration Display 2)

■ Current Output Characteristics (YS1360)
The current output of the YS1360 manual setter for SV setting provides a forced full-closing function for valves (final control elements).
When a signal decreases, if an output signal becomes -6% (equivalent to 3.04 mA) or less, the current output decreases abruptly, reaching -20% (equivalent to 0.8 mA).
When a signal increases, if the output signal reaches -5% or more, the current output is released from -20% (equivalent to 0.8 mA) and returns to -5% (equivalent to 3.2 mA).
When the output signal is 100% or more, the current output increases linearly and is limited at approximately 106.25% (21 mA).



■ High and Low Limiters for Manipulated Output Signals (YS1360)

In mode C (Analog cascade or Computer cascade), limiters can be set for manipulated output signals. (This feature does not function in mode M.)
High limit, MH1: -6.3 to 106.3%
Low limit, ML1: -6.3 to 106.3%

Main Parameter Functions

Main Functions	Reference Destination
Square root extraction	1.2.1 Square Root Extraction (Low Cutoff Adjustable)
Digital input function	1.5 Setting Digital Input Function

Other Function

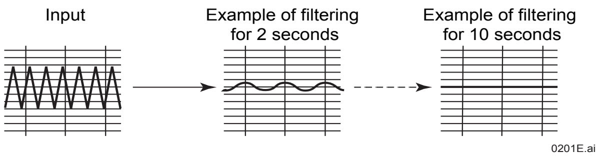
Main Functions	Reference Destination
Display function	Chapter2, Display/Security Functions
Communication function	YS1000 Series Communication Interface User's Manual

1.2 Compensating or Computing Process Variables

1.2.1 Input Filter (First-order Lag Operation)

Description

First-order lag operation is performed on PV input. This function is used if there are significant variations in the display value such as the presence of noise. The greater the time constant, the stronger the filtering function.



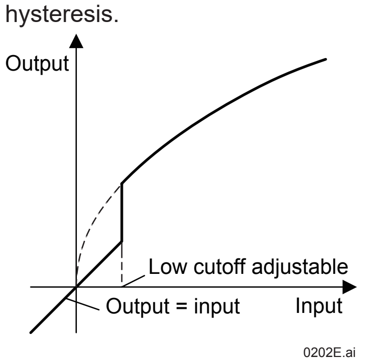
Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
PLG1	First-order lag time constant for PV	0.0 to 800.0 sec	Tuning Display > [**SETTING**] (Setting Display)

1.2.2 Square Root Extraction (Low Cutoff Adjustable)

Description

Square root extraction is performed on a process variable (PV). The input and output characteristics are that input = output at the low cutoff point or below. There is no hysteresis.



Square Root Extraction Characteristics

Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
PSR1	Square root extraction for PV1	OFF, ON	Tuning Display > Engineering Display > [**CONFIG3**] (Configuration Display 3)
PLC1	Square root extraction low cutoff setpoint for PV	0.0 to 100.0%	Tuning Display > [**SETTING**] (Setting Display)

1.3 Changing the Valve Direction (YS1360)

Description

The valve direction is used to decide "C-O" or "O-C" displayed in LCD.

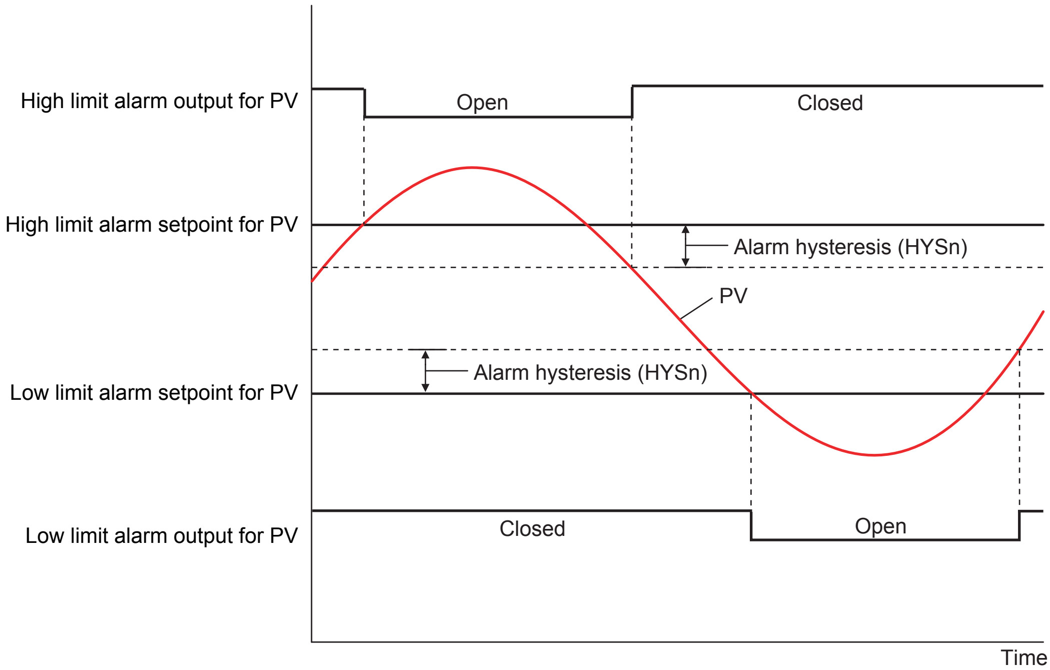
Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
VDIR1	Valve direction	C–O: Manipulated output variable 0% = C 100% = O O–C: Manipulated output variable 0% = O 100% = C	Tuning Display > Engineering Display > [**CONFIG2**] (Configuration Display 2)

1.4 Setting Alarm Output Hysteresis

Description

Setting hysteresis (differential gap) to alarm action prevents chattering in digital output.



For an example in the figure above, the contact type is such that the contact opens if an event occurs (factory default).

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Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
HYS1	Alarm hysteresis	Equivalent to 0.0 to 20.0% in the engineering unit	Tuning Display > [**SETTING**] (Setting Display)

1.5 Setting Digital Input Function

Description

Changing action of Operation mode by digital input.
When using this in the analog cascade mode (CMOD1=CAS), the operation mode can be switched through digital input.

<table>
<tr><th>Operation Mode Display Lamp</th><th>DI1F Parameter</th><th>DI1D Parameter</th><th>Digital Input</th><th>LOOP Display Control Status Display</th><th>Operation Mode</th><th>Operation Mode Status Output</th></tr>
<tr><td>M lamp lights</td><td>- or E-MAN</td><td>OPN or CLS</td><td>OPEN or CLOSE</td><td>No display</td><td>M</td><td>OPEN</td></tr>
<tr><td>C lamp lights</td><td>–</td><td>OPN or CLS</td><td>OPEN or CLOSE</td><td>CAS</td><td>C</td><td>CLOSE</td></tr>
<tr><td>C lamp lights and M lamp blinks</td><td rowspan="4">E-MAN</td><td rowspan="2">OPN</td><td>OPEN</td><td>CAS,
EXT-MAN</td><td>M</td><td>OPEN</td></tr>
<tr><td rowspan="2">C lamp lights</td><td>CLOSE</td><td>CAS</td><td>C</td><td>CLOSE</td></tr>
<tr><td rowspan="2">CLS</td><td>OPEN</td><td>CAS</td><td>C</td><td>CLOSE</td></tr>
<tr><td>C lamp lights and M lamp blinks</td><td>CLOSE</td><td>CAS,
EXT-MAN</td><td>M</td><td>OPEN</td></tr>
</table>

Switching to manual control (TR-MAN)

When a digital input signal (DI) is switched from "Close" to "Open", the operation mode switches to manual (MAN) mode. The M mode key lights in yellow in the Operation Display.
When a digital input signal (DI) is switched from "Open" to "Close", the operation does not change.
(When the contact type is OPN)

Switching to cascade control (TR-CAS)

When a digital input signal (DI) is switched from "Close" to "Open", the operation mode switches to cascade (CAS) mode. The C mode key lights in green in the Operation Display.
When a digital input signal (DI) is switched from "Open" to "Close", the operation does not change.
(When the contact type is OPN)

All event elimination (TR-EVT.C)

When a digital input signal (DI) is switched from "Close" to "Open", all the displayed events are cleared. For all other cases, events are displayed when the display conditions are met.
After the event display is cleared, if the digital input signal (DI) remains "Open", when an event display condition is met, the corresponding event (1 to 5) is displayed.
(When the contact type is OPN)

If any front panel key is pressed once while the LCD backlight is OFF, it will be turned ON again.

Setting Details

<table>
<tr><th>Parameter</th><th>Name</th><th>Setting Range</th><th>Display Transition and Display Title</th></tr>
<tr><td>DI1F</td><td>DI1 function selection</td><td>NONE: No function
E-MAN: Switching to manual mode (status)
TR-MAN: Switching to manual mode
TR-CAS: Switching to cascade mode
TR-EVT.C: All event OFF</td><td rowspan="4">Tuning Display >
Engineering Display >
[CONFIG3] (Configuration Display 3)</td></tr>
<tr><td>DI2F</td><td>DI2 function selection</td><td>NONE: No function
TR-MAN: Switching to manual mode
TR-CAS: Switching to cascade mode
LCD-OFF: LCD backlight OFF
TR-EVT.C: All event OFF</td></tr>
<tr><td>DI1D</td><td>DI1 contact type</td><td rowspan="2">OPN: Function is available when the contact is open.
CLS: Function is available when the contact is closed.</td></tr>
<tr><td>DI2D</td><td>DI2 contact type</td></tr>
</table>

1.6 Setting Digital Output Function

Description

There are the follwing three digital outputs.
DO1: Hig limit alarm output, current output open status output or no function
DO2: Low limit alarm output, current output open status output or no function
DO4: C/M status output, current output open status output or no function

Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
DO1F	DO1 function selection	NONE: No function assigned PH1: High limit alarm for PV1 OOP: Current output open	Tuning Display > Engineering Display > [**CONFIG3**] (Configuration Display 3)
DO2F	DO2 function selection	NONE: No function assigned PL1: Low limit alarm for PV1 OOP: Current output open	
DO4F	DO4 function selection	NONE: No function assigned CAS: Cascade mode OOP: Current output open	
DO1D	DO1 Contact type	OPN: When the event occurs, the contact is open. CLS: When the event occurs, the contact is closed.	
DO2D	DO2 Contact type		
DO4D	DO4 Contact type		

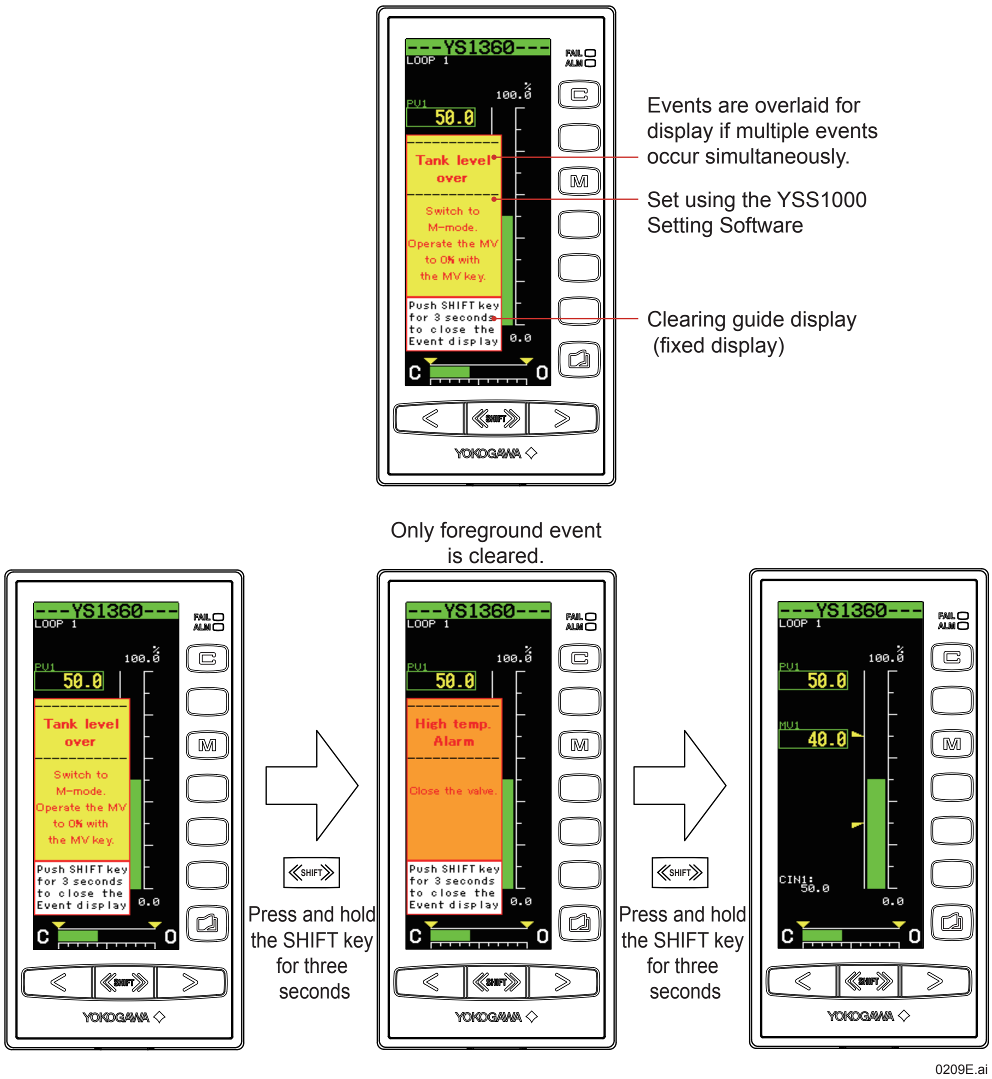
1.7 Using the Event Function

1.7.1 Displaying Messages (Event Display Function (Settable Only in YSS1000))

Description

The event display function displays a message (in Japanese, English, or Chinese) on the Operation Display if an event such as an abnormality occurs, providing guidance for the operator. A maximum of five events can be set using the YSS1000 Setting Software (sold separately). Events will be displayed on all Operation Displays. The display area is 200 × 80 dots, with 50 × 80 dots of that used to display a clearing guide.
If multiple events occur simultaneously, they are displayed in an overlaid condition. The order of priority is (high) event 1 > 2 > 3 > 4 > 5 (low), and the event with the highest priority is displayed on top. The event being displayed can be cleared by pressing and holding the SHIFT key for three seconds, contact input (DI), or communication. Moreover, events can be redisplayed on Alarm Display.

► For how to redisplay events: see "Monitoring and Operating the Alarm Display" in the YS1350 Manual Setter for SV Setting/YS1360 Manual Setter for MV Setting Operation Guide.



Event flag

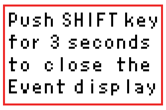
The status register is the flag to indicate an event. An event is displayed if the status register changes from 0 to 1 or from 1 to 0.
Process alarm such as a high limit alarm and operation mode can be registered as a flag.

Simulation Display

After downloading the event display data from YSS1000, the event can be displayed forcibly. Simulation display can be set when operation is stopped.
To cancel simulation display:
(1) Turn OFF simulation display on YSS1000.
(2) Change the operation stop status from stopped to operating.
(3) Turn the power supply OFF and then ON.

Deletion Guide Display Language

English



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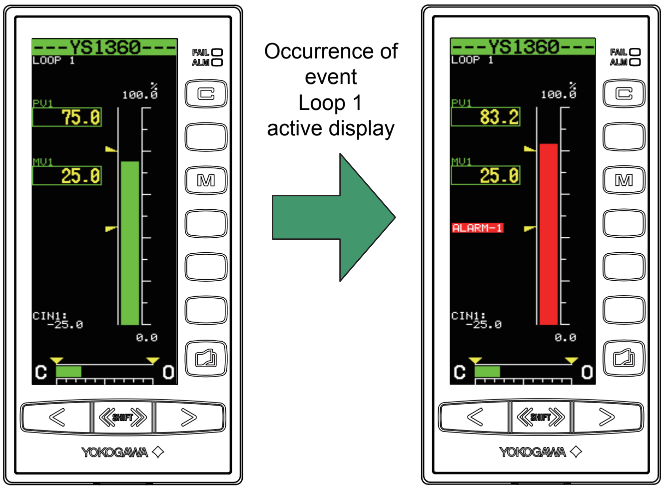
1.7.2 Changing the PV Bar Display Color in the Event of a Process Alarm (Active Display Function)

Description

The active color display function changes the colors of the PV bar on the LOOP Display to red to inform the operator of instrument abnormality. Active color display can be set on a loop basis and items can be selected from each loop's process alarms (see the table below).



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Setting Details

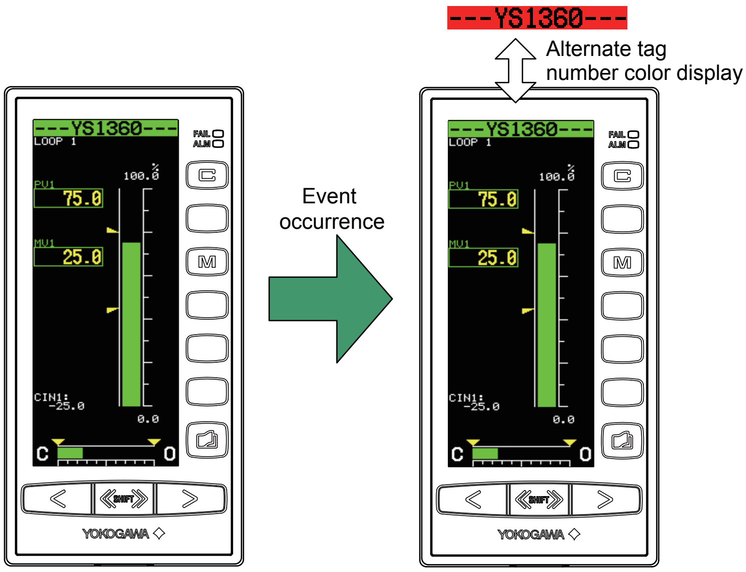
Parameter	Name	Setting Range	Display Transition and Display Title
ACTD1	Active color display selection 1	OFF: None PH1: High limit alarm setpoint for PV 1 PL1: Low limit alarm setpoint for PV 1 1-ALM: Logical OR of all loop 1 alarms	Tuning Display > Engineering Display > [**DISPLAY**] (Setting Display for Operation Display)

1.7.3 Operator Notification Using Tag Number Display (Alternate Tag Number Color Display Function)

Description

In the event of an ALM lamp lighting, the alternate tag number color display function notifies the operator of instrument abnormality by alternating the color of the tag number background back and red.

► For lighting of ALM lamp: see "How to Take Actions if the ALM Lamp or FAIL Lamp Lights up " in the YS1350 Manual Setter for SV Setting/YS1360 Manual Setter for MV Setting Operation Guide.



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Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
TAGAL	Color inversion of tag number	OFF: Disabled ON: Enabled	Tuning Display > Engineering Display > [**DISPLAY**] (Setting Display for Operation Display)

1.8 Setting the Display for When Measurement Input Is Not Wired

Description

When measurement input 1 is not used, it is possible to prevent (unnecessary) alarms (warnings) from occurring. It is also not necessary to wire the input.

When measurement input 1 input specification (PV1IN) is set to "-"

PV1 displays SCL1 (scale 0%) regardless of the input (1 to 5 V) to the X1 terminal.
The value (1 to 5 V) applied to the X1 terminal is displayed at X1 of [Tuning Display] > [I/O DATA].
When the X1 terminal is open, the ALM lamp does not light.
When the X1 terminal is open, X1 does not appear under [Alarm Display] > [SYSTEM].

When measurement input 1 input specification (PV1IN) is set to "X1"

The value (1 to 5 V) applied to the X1 terminal is displayed at PV1.
The value (1 to 5 V) applied to the X1 terminal is displayed at X1 of [Tuning Display] > [I/O DATA].
When the X1 terminal is open, the ALM lamp lights.
When the X1 terminal is open, X1 appears under [Alarm Display] > [SYSTEM].

If an A/D converter error (A/D) occurs, the input fails regardless of the measurement input 1 input specification (PV1IN) setting.

Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
PV1IN	PV1 input specification	-: Not use X1: Use PV1	Tuning Display > Engineering Display > [**CONFIG2**] (Configuration Display 2)

2.1 Display Function

2.1.1 Setting Visible/Invisible Status of the Operation Display

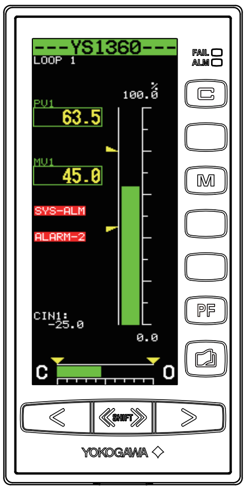
Description

There are five types of Operation Displays. Displays you chose not to display can be made invisible.
LOOP1 Display is always displayed even if you set "OFF" for all lines.

Operation Display Name	YS1350/YS1360
LOOP1 (LOOP 1 Display)	✓
MTR1 (METER 1 Display)	✓
TRND1 (TREND 1 Display)	✓
TRND3 (TREND 3 Display)	✓
ALRM (Alarm Display)	✓

Legend ✓: Visible

LOOP 1 Display



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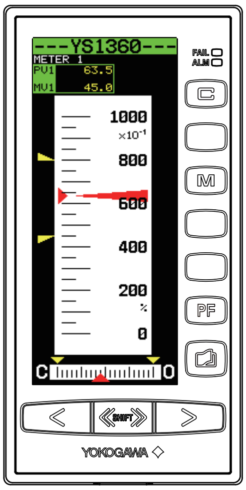
METER 1 Display



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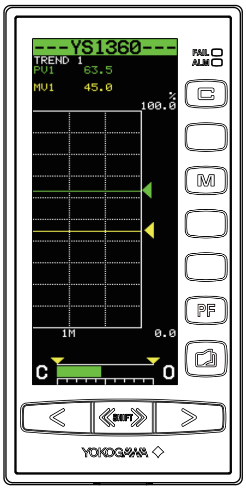
TREND 1 Display



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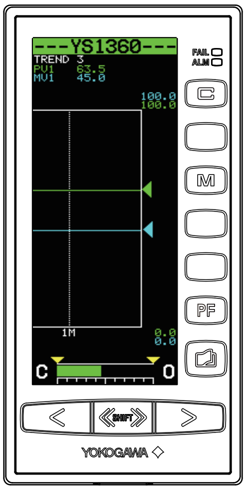
TREND 3 Display



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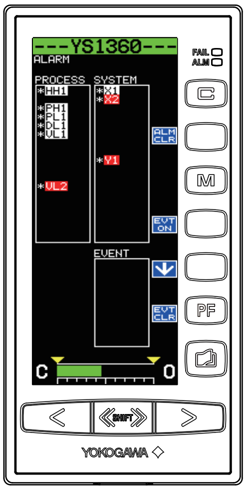
Alarm Display



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Examples of YS1360

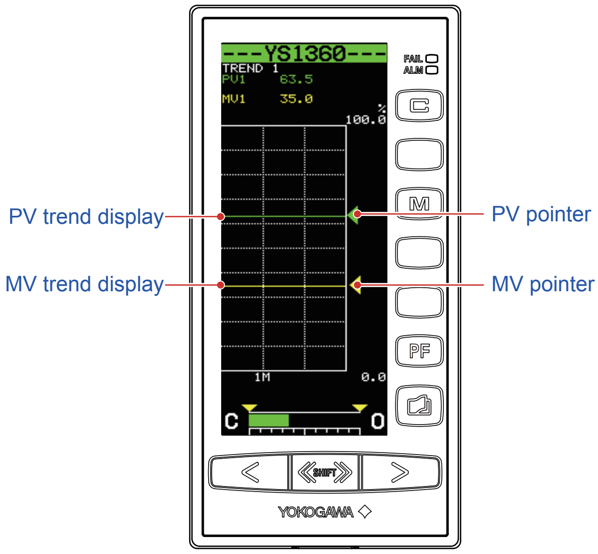
Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
LOOP1	LOOP 1 Display ON/OFF	OFF: Invisible ON: Visible	Tuning Display > Engineering Display > [**CONFIG1**] (Configuration Display 1)
MTR1	METER 1 Display ON/OFF		
TRND1	TREND 1 Display ON/OFF		
TRND3	TREND 3 Display ON/OFF		
ALARM	Alarm Display ON/OFF		

2.1.2 Setting Visible/Invisible Status of TREND Display Data

Description

PV trend, SV trend, and MV trends displayed on TREND 1 Display can be set to be visible or invisible. It is possible to select the trends that are to be displayed, making only the data necessary to be monitored visible on the Operation Display.



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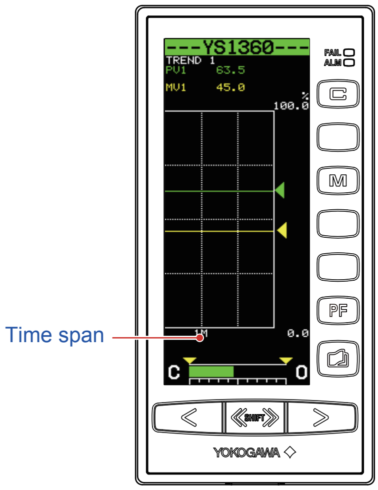
Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
TR1PV	PV1 trend ON/OFF for TREND 1 Display	OFF: Invisible ON: Visible	Tuning Display > Engineering Display > [**DISPLAY**] (Setting Display for Operation Display)
TR1SV	SV1 trend ON/OFF for TREND 1 Display (Only for YS1350)		
TR1MV	MV1 trend ON/OFF for TREND 1 Display (Only for YS1360)		

2.1.3 Changing the Time Span of TREND Displays

Description

The time span of trends displayed on the TREND 1, and TREND 3 Displays can be set.



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Setting Details

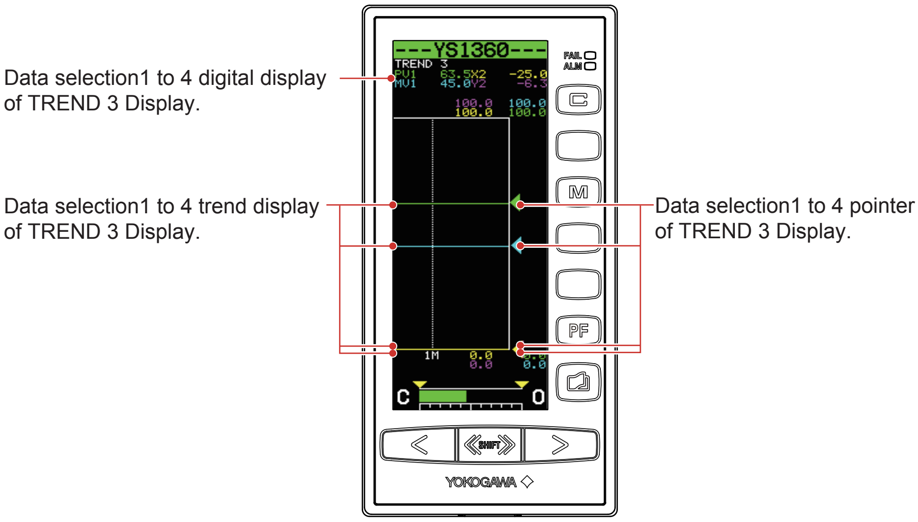
Parameter	Name	Setting Range	Display Transition and Display Title
TRDT1	TREND 1 Display time span	1M: 1 minute 5M: 5 minutes 10M: 10 minutes 30M: 30 minutes 1H: 1 hour 5H: 5 hours 10H: 10 hours 30H: 30 hours	Tuning Display > Engineering Display > [**DISPLAY**] (Setting Display for Operation Display)
TRDT3	TREND 3 Display time span		

2.1.4 Setting Display Data on the TREND 3 Display

Description

Display data shown on the TREND 3 Display can be set arbitrarily. A maximum of four data items can be set. This enables the necessary data to be monitored on the Operation Display.

TREND Display 3



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Setting Details

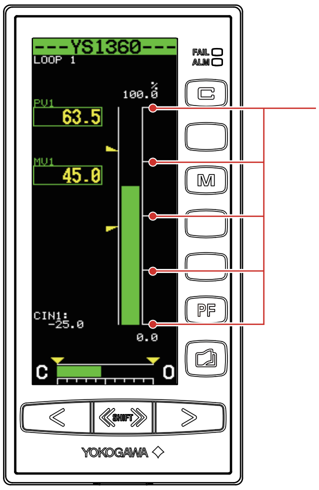
Parameter	Name	Setting Range (*1)	Display Transition and Display Title
TRDS1	Data selection 1 for TREND 3 Display	OFF: None PV1: Process variable 1 SV1: Setpoint value 1 (YS1350) MV1: Manipulated output variable 1 (YS1360) X1: Analog input 1 X2: Analog input 2 Y1: Analog output 1 (YS1360) Y2: Analog output 2	Tuning Display > Engineering Display > [**DISPLAY**] (Setting Display for Operation Display)
TRDS2	Data selection 2 for TREND 3 Display		
TRDS3	Data selection 3 for TREND 3 Display		
TRDS4	Data selection 4 for TREND 3 Display		

*1: X1 to X2 and Y1 to Y2 are values input to or output from the YS1350/YS1360 terminal block.

2.1.5 Changing Scale Divisions on the LOOP, TREND, and DUAL Displays

Description

The provision of scale divisions is relevant to the LOOP, TREND, and DUAL Displays. For scale divisions on the METER Displays, see 2.1.6 Automatic Scale Divisions/Making Scale Values More Legible on the METER Displays



Scale division
The figure on the left is an example of four scale divisions.

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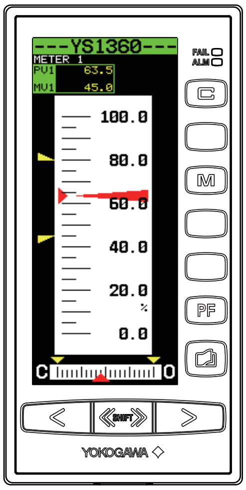
Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
SCDV1	Scale division	1, 2, 4, 5, 7, 10, 14, 20	Tuning Display > Engineering Display > [**CONFIG2**] (Configuration Display 2)

2.1.6 Automatic Scale Divisions/Making Scale Values More Legible on the METER Display

Description

For scales on the METER Displays, scale divisions are automatically provided based on the values set to the scale between 0% and 100% values.
The number of scale divisions is from a minimum of 11 divisions to a maximum of 20 divisions.



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Scale numbers displayed on the METER Display are also automatically determined from the scale's 0% to 100% values in the same way as the scale divisions. To improve the legibility of the scale numbers, they can be displayed to the power of 10.

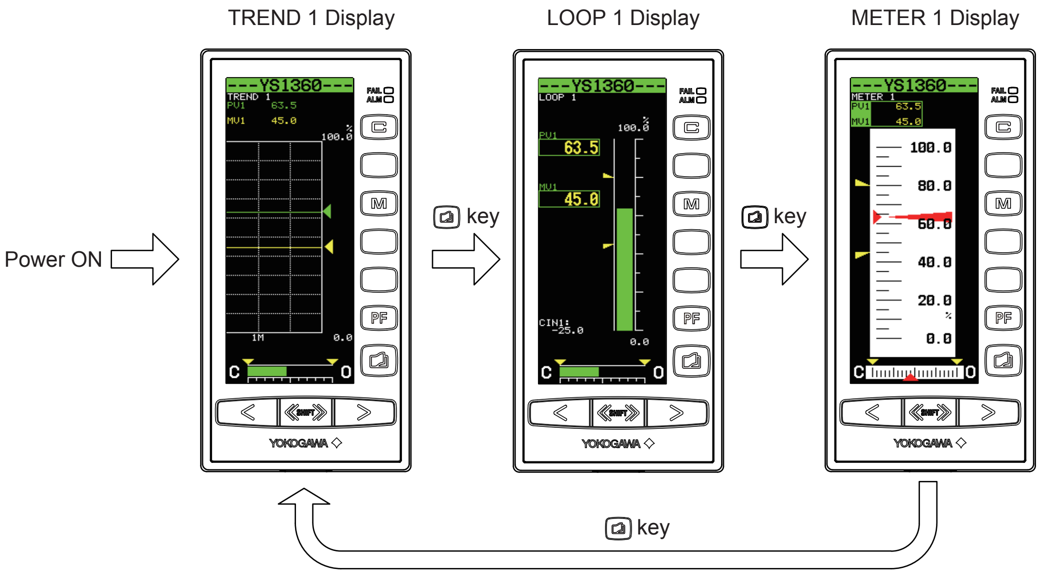
Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
MTMG1	10-exponential scale factor for METER 1 Display	AUTO, 10^-5,10^-4, 10^-3,10^-2,10^-1,10^0, 10^1,10^2,10^3,10^4, 10^5,	Tuning Display > Engineering Display > [**DISPLAY**] (Setting Display for Operation Display)

2.1.7 Selecting the Operation Display to be Displayed First at Power ON

Description

The Operation Display to be displayed first when the power is turned ON can be set. The figure below shows an example of displaying the TREND 1 Display first.



Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
FDSP	Power-on initial display	LOOP1: LOOP 1 Display MTR1: METER 1 Display TRND1: TREND 1 Display TRND3: TREND 3 Display ALARM: Alarm Display	Tuning Display > Engineering Display > **[CONFIG1]** (Configuration Display 1)

2.1.8 Turning the LCD Backlight ON/OFF

Description

The backlight auto-off function makes it possible to turn off the LCD backlight in cases where the YS1000 is installed in locations where the display is not usually seen or if it desired to turn it off at night. This is an energy-saving feature that extends the life of the display unit.

The LCD backlight can be turned ON/OFF using the following means:
(1)Front panel keys
(2)OFF timer setting
(3)Digital input
(4)Communication

There is no priority order to the methods (1) to (4). The backlight OFF function retains the status effected when the LCD was operated last.
Note that if the FAIL or ALM lamps are lit, or if an event is displayed, the backlight will light up even if the backlight has been set to OFF.

(1)Front panel key
If any key is pressed once while the LCD backlight is OFF, it will be turned ON. However, it cannot be turned OFF by keystrokes. To set the backlight condition to OFF using keystrokes, employ method (2).

(2)OFF timer setting
With the LCD backlight auto-off timer turned ON, the LCD backlight will be turned OFF when there has been no keystroke operation for 30 minutes.

(3)Digital input
If digital input to which the backlight OFF function has been assigned changes from open to close status, the backlight is turned OFF. If it changes from close to open status, the backlight is turned ON.
► For assigning the backlight OFF function to digital input: see Chapter 1, Input/output and Auxiliary Functions, in this manual.

(4)Communication
The LCD backlight ON/OFF condition can be checked and set using a CFL flag (communication register). Write “0” or “1” to the CFL flag through communication to (0) to turn the backlight ON or to (1) to turn it OFF.
► For backlight OFF function through communication: see YS1000 Series Communication Interface User’s Manual.

Setting Details

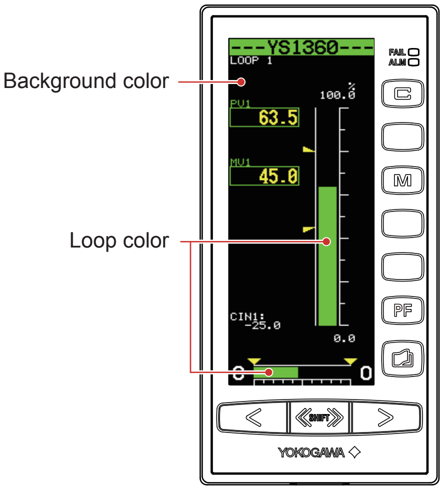
Parameter	Name	Setting Range	Display Transition and Display Title
ECO	LCD backlight auto-off timer	OFF: Timer function OFF ON : Timer function ON (Off timer: 30 min)	Tuning Display > Engineering Display > [**LCD**] (LCD Setting Display)

2.1.9 Changing the Background and Loop Colors

Description

The background color of the Operation Display, and the color of the PV and MV bars of the LOOP Displays and the DUAL Display can be changed.
Setting the Operation Display’s background color to black causes the color of the lettering to be white; setting the background color to white causes the color of the lettering to be black.

LOOP 1 Display



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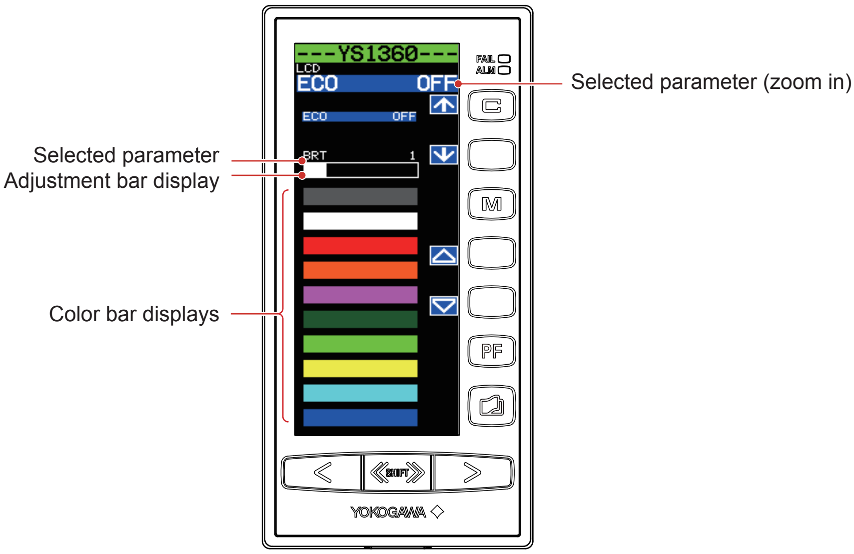
Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
BKCL	Background color selection	BLACK WHITE BLUE	Tuning Display > Engineering Display > [**DISPLAY**] (Setting Display for Operation Display)
LP1C	Loop 1 color selection	GREEN AQUA PINK ORANGE	

2.1.10 Adjusting LCD Brightness

Description

The brightness of the LCD can be adjusted.
The adjustment bar display shows brightness adjustment values in bar format to indicate the current value with respect to the settable range.



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Setting Details

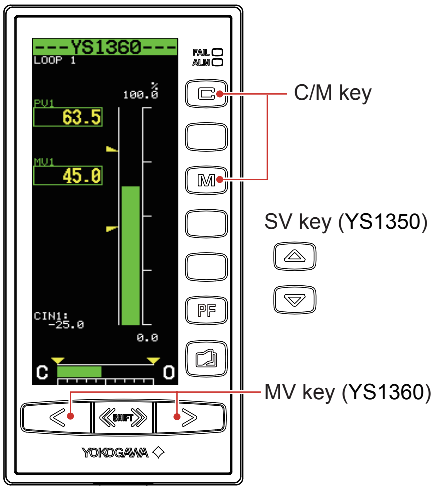
Parameter	Name	Setting Range	Display Transition and Display Title
BRT	LCD brightness adjustment	0 to 5	Tuning Display > Engineering Display > [**LCD**] (LCD Setting Display)

2.2 Security Function

2.2.1 Setting/Releasing Keylock

Description

Keys can be locked to prevent inadvertent operation on the Operation Display.



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Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
CAMLK	Keylock for C/M mode change	UNLOCK: Keys unlocked LOCK: Keys locked	Tuning Display > Engineering Display > [**CONFIG1**] (Configuration Display 1)
SVLK	Keylock for SV change (YS1350)		
MVLK	Keylock for MV change (YS1360)		

2.2.2 Inhibiting/Enabling Parameter Change

Description

Setpoints for the tuning parameters or engineering parameters can be locked to prevent them from being inadvertently changed. When they are locked, the parameter setting increase key [△] and decrease key [▽]displays are erased from each setting display.
The password for locking them is a 4-digit numeral, which is not set at factory shipment. When the password is not set or is cleared, [SET PASSWORD] and [UNLOCK] are displayed. When the password is set, [ENT PASSWORD] and [LOCK] are displayed.

Software key function

The software key function enables the front panel's operation keys to function as keys displayed on the LCD display.

- [SET] software key: Password setting key
  Press this key to set a password.
- [ENT] software key: Password entry/cancellation key
  Press this key to cancel the password setting.
- [→] key: Cursor movement key
  Moves the cursor position to the right when setting or entering a password.
- [△] software key: Number increase key
  Increases numbers. Numbers change from 0 to 9 cyclically.
- [▽] software key: Number decrease key
  Decreases numbers. Numbers change from 0 to 9 cyclically.

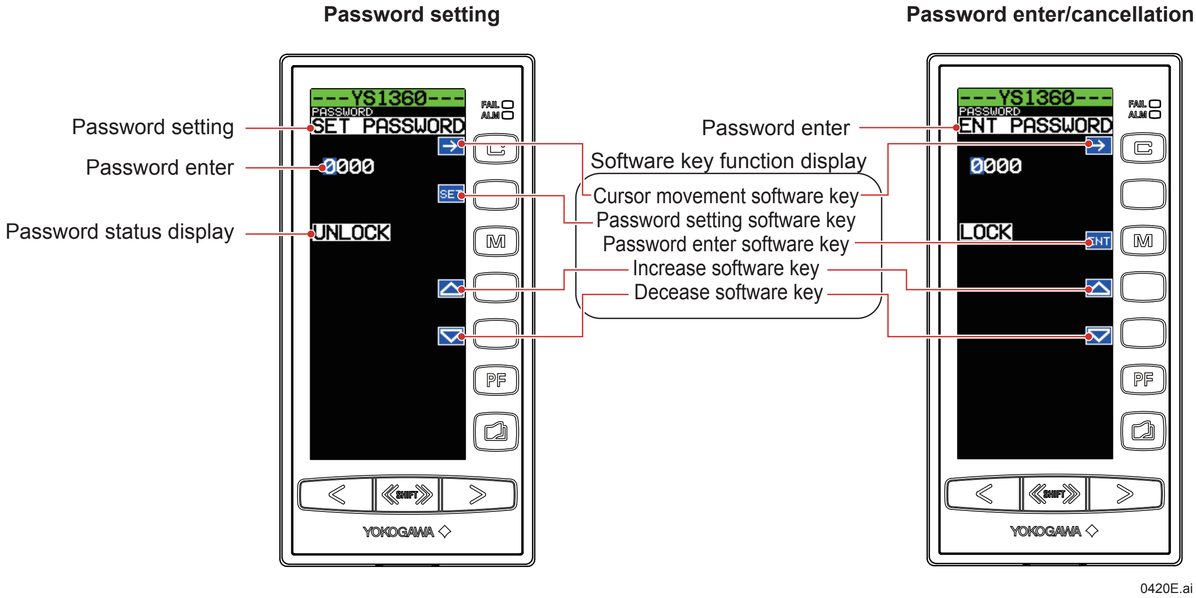
Operating the Password Setting Display

1. Setting a password (setting a password to prevent parameter changes)
   (1)Open the Password Setting Display. [SET PASSWORD] and [UNLOCK] are displayed.
   (2)Press the [SET] software key. This causes the password [0000] to appear.
   (3)Using the [→] software key (digit movement) and the [△] (increase) or [▽] (decrease) software keys, set the password.
   (4)Press the [SET] software key. This changes the password background color.
   (5)Press the [SET] software key again. This erases the password, causing [ENT PASSWORD] and [LOCK] to appear. At the instant the password is set, the [SET] software key display is erased, replaced by the [ENT] software key instead.

2. Entering/canceling a password (entering a password to the instrument to which the password has been set, to enable parameter changes)
   (1)Open the Password Setting Display. [ENT PASSWORD] and [LOCK] are displayed.
   (2)Press the [ENT] software key. This causes the password [0000] to appear.
   (3)Using the [→] software key (digit movement) and the [△] (increase) or [▽] (decrease) software keys, enter the password that has been set.
   (4)Press the [ENT] software key. This changes the password background color.
   (5)Press the [ENT] software key again. This erases the password if the password entered agrees with the one that was set, causing [SET PASSWORD] and [UNLOCK] to appear. This brings about a status allowing parameters to be changed. If the password entered does not agree with the one that has been set, return to step (3).

Setting/Entry Display

Operation Display > [SHIFT]+[ ] keys (to the Tuning Menu Display) > [SHIFT]+[ ] keys (to the Engineering Menu Display) > [PASSWORD] software key (Password Setting Display)



3.1 List of Direct Input Specifications and Basic Operations

In the SC Setting Display, communication with the direct input cards, optional specifications, is made to set or adjust the input specifications.
Examples of sensor type setting, burnout setting, input zero adjustment, and span adjustment procedures are described in sections 3.2 to 3.4. Other items are also set in the same way. The table below shows a list of setting items.

No.	Name Display	Description	Model-basis Data Display				
			/A01	/A02	/A03	/A04	/A08
01	MODEL	Model	EM1*C	ET5*C	ER5*C	ES1*C	EP3*A
02	TAG NO.	Tag number	16 alphanumeric characters				
03	SELF CHK	Self-check results	GOOD or ERROR				
A00	DISPLAY	Display item	--	--	--	--	--
A01	INPUT	Input value	[ ][ ][ ].[ ][ ]mV	[ ][ ][ ][ ].[ ]degC	[ ][ ][ ].[ ]degC	[ ][ ][ ][ ][ ].[ ] OHM	[ ][ ][ ][ ]Hz
A02	OUTPUT	Output value	[ ][ ][ ].[ ]%				
A03	STATUS	Status	FF				
A04	REV NO.	Revision number	n.000 (n: Revision number)				
B00	SET	Setting item	--	--	--	--	--
B01	TAG NO1	Tag number 1	8 alphanumeric characters (8 first-half characters of a tag number)				
B02	TAG NO2	Tag number 2	8 alphanumeric characters (8 latter half characters of a tag number)				
B03	COMMENT1	Comment 1	8 alphanumeric characters (8 first-half characters of comment)				
B04	COMMENT2	Comment 2	8 alphanumeric characters (8 latter half characters of comment)				
B05	INP TYPE	ER5 input type	--	--	PT/JPT/ PT100-90/ PT50 (Note 1)	--	--
B06	INP TYPE	ET5 input type	--	B/E/J/K/T/R/S/N	--	--	--
B07	LOW CUT	Low cutoff	--	--	--	--	[ ][ ][ ][ ]Hz (Note 5)
B08	RESIST	ES1 total resistance	--	--	--	[ ][ ][ ][ ][ ].[ ] OHM	--
B09	UNIT	Temperature unit (Note 8)	--	degC/degF/K	degC/degF/K	--	--
B10	ZERO	Zero point	[ ][ ][ ].[ ][ ]mV	[ ][ ][ ][ ].[ ]degC	[ ][ ][ ].[ ]degC	[ ][ ][ ][ ][ ].[ ] OHM	[ ][ ][ ][ ]Hz (Note 5)
B11	SPAN	Span (Note 2)	[ ][ ][ ].[ ][ ]mV	[ ][ ][ ][ ].[ ]degC	[ ][ ][ ].[ ]degC	[ ][ ][ ][ ][ ].[ ] OHM (Note 4)	[ ][ ][ ][ ]Hz (Note 5)
B12	BURN OUT	Burnout	OFF/UP/DOWN	OFF/UP/DOWN	OFF/UP/DOWN	OFF/UP/DOWN	--
C00	ADJUST	Adjustment item	--	--	--	--	--
C01	OUT 0%	0% output correction (Note 7)	±10.00				
C02	OUT 100%	100% output correction (Note 7)	±10.00				
C03	WIRING R	Burnout correction (Note 3)	EXECUTE/RESET	EXECUTE/RESET	--	--	--
C04	ZERO ADJ	Input zero adjustment (Note 6)	[ ][ ][ ].[ ][ ][ ]mV RST/INC/DEC	[ ][ ][ ].[ ][ ][ ]mV RST/INC/DEC	[ ][ ][ ].[ ][ ][ ] OHM RST/INC/DEC	--	--
C05	SPAN ADJ	Input span adjustment (Note 6)	[ ][ ][ ].[ ][ ][ ]mV RST/INC/DEC	[ ][ ][ ].[ ][ ][ ]mV RST/INC/DEC	[ ][ ][ ].[ ][ ][ ] OHM RST/INC/DEC	--	--
C06	ZERO ADJ	Input zero adjustment (Note 6)	--	--	--	[ ][ ][ ].[ ][ ][ ] OHM	--
C07	SPAN ADJ	Input span adjustment (Note 6)	--	--	--	[ ][ ][ ].[ ][ ][ ] OHM	--

Note 1: PT=Pt100 (IPTS-68: JIS'89), JPT=JPt100 (JIS'89), PT100-90=Pt100 (ITS-90: JIS'97), PT50=Pt50 (JIS'81)
Note 2: Measurable data is within the range stated in the standard specifications.
Note 3: Burnout correction is the function of correcting an error caused by the burnout current produced if an external conductor resistance is large.
Note 4: Up to 30 kΩ is possible, but the standard specifications indicate 100 to 2000 Ω in.
Note 5: Set in four significant digits or less. However, 10000 Hz can be set for span.
Note 6: Input zero adjustment and input span adjustment are to make input adjustments of each direct input card. /A01, /A02, and /A03 enable the adjustment of the offset and gain of the A/D converter. Select [INC] or [DEC] using the [△] or [▽] software keys and press the [ENT] software key twice to make an adjustment for each selection. Also, to reset adjustments select [RST] and press the [ENT] software key twice. Zero and span are re-set in /A04. In other words, making a zero adjustment with an input set to 0% and a span adjustment with the input set to 100% (pressing the [ENT] software key twice) causes the zero (B10) and span (B11) values to be re-set automatically.
Note 7: Output correction is for adjusting the D/A converter (1 to 5 V output) of each direct input card. 0% output correction and 100% output correction enable offset and gain to be adjusted respectively. Set a value in the range of ±10.00% and press the [ENT] software key twice. This causes the D/A converter to enter a status in which the set value-added 0% output value or the 100% output value is continuously output. To exit this status, display another parameter once on the SC Setting Display, or turn the power supply OFF and then ON.
Note 8: If optional code /DF is specified, Fahrenheit tempertature range can be used for direct input range in addition to centigrade temperature range.

Basic Operations

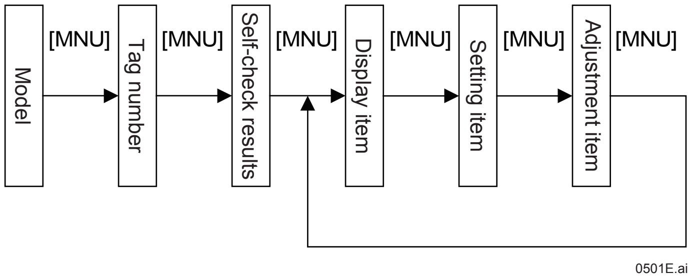
1) Erroneous setting prevention function
To prevent inadvertent operation, no parameter is selected (highlight displayed) immediately after switching to the SC Setting Display. Press the [→] software key to select parameter [SET] (parameter setting enable/disable) from the top line of the display.
2) Setting-enable parameter operation
[SET] on the top line is the parameter for enabling SC maintenance communication. This parameter is in [INHB] (setting inhibited) immediately after switching to the SC Setting Display. SC maintenance communication cannot be accomplished unless this parameter is set to [ENBL] (setting enabled). To do so, select the [SET] parameter and then press the [△] software key to change the setting from [INHB] to [ENBL]. At the same time, [STOP] appears at the upper right of the display. When the parameter is set to [ENBL], the operation mode is forced to switch to manual control (M mode), retaining the manipulated output and alarm output. Moreover, switching to another display causes this parameter to return to [INHB] automatically.
3) Software keys
   - [MNU] software key: Menu change key
     Each time this key is pressed, YS1350/YS1360 communicates with the SC to read out and display the SC menus.
   - [PRM] software key: Setting item change key
     Each time this key is pressed, YS1350/YS1360 communicates with the SC to read out and display the SC setting items.
   - [→] software key: Cursor movement key
     When the data type is alphanumeric characters, this key moves an highlight displayed digit to the right. In this case, should the highlight display be at the rightmost digit, it then moves to the leftmost digit.
   - [△] software key: Data increase key
     Increases data. Data changes cyclically.
   - [▽] software key: Data decrease key
     Decreases data. Data changes cyclically.
   - [ENT] software key: Enter key
     Writes data to the SC. Writing is done in the following two steps:
     (1)Press the [ENT] software key. This causes the background color of all communication data to be highlight displayed.

(2)Press the [ENT] software key again. This writes the data to the SC card, causing the display to return to normal display. If any key other than the [ENT] software key is pressed, the communication data returns to normal display without being written to the SC.

4) SC setting operation

- SC setting is made as follows:

(1)Selecting the SC menu
Press the [MNU] software key to read out and display the SC menus. Each time the [MNU] software key is pressed, the model, tag number, and self-check results are displayed in turn. Further pressing the [MNU] key causes the display item, setting item, and adjustment items to be changed and displayed cyclically.



(2)To select the display item, setting item, or adjustment item, further select a setting item.
Press the [PRM] software key to read out and display the SC setting item. Each time the [PRM] software key is pressed, the setting items are displayed in turn. The parameters to be displayed change depending on the SC card; see the items in the list of displayed items. Press the [PRM] software key until the setting item you wish to set appears.

(3)Using the [→] software key (cursor movement) and the [△] (data increase) or [▽] (data decrease) software keys, set the setting item.

(4)Press the [ENT] software key. This causes the setting item to be highlight displayed.

(5)Press the [ENT] software key again. This writes data to the SC card, causing the display to return to the normal display.
Pressing any key other than the [ENT] software key causes the display to return to the normal display without writing data.

Communication Status Display

Comment Display	Status
COMMUNICATING	Communicating with an SC card (normal communication)
COMMUN. ERR	Communications error
OPERATION ERR	Incorrect data sending
COMMAND ERR	Incorrect command reception

3.2 Setting Sensor Type

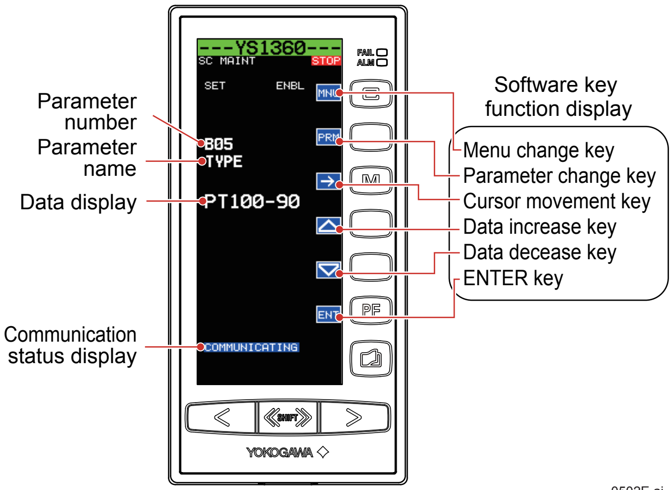
Description

The sensor type can be set as for thermocouple and RTD cards.

Setting Details

Operation Display > [SHIFT]+[◰] keys (to the Tuning Menu Display) > [SHIFT]+[◰] keys (to the Engineering Menu Display) > [**SC MAINT**] software key (Input Specification Setting Display)

Engineering Display



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Setting Procedure

(example of setting an RTD type)

(1)Press the [→] software key to display [SET INHB] in highlight.

(2)Press the [△] software key to display [SET ENBL].

(3)Press the [MNU] software key to display
01
MODEL
Directly input the card model (e.g., ER5*C)

(4)Press the [MNU] key several times to display
B00
SET

(5)Press the [PRM] software key several times to display the left display.

(6)Press the [△] or [▽] software keys to change the data display section.

(7)Press the [ENT] software key to display the data display section in highlight.

(8)Press the [ENT] software key again to accept the setting.

Setting completed.

3.3 Setting Burnout

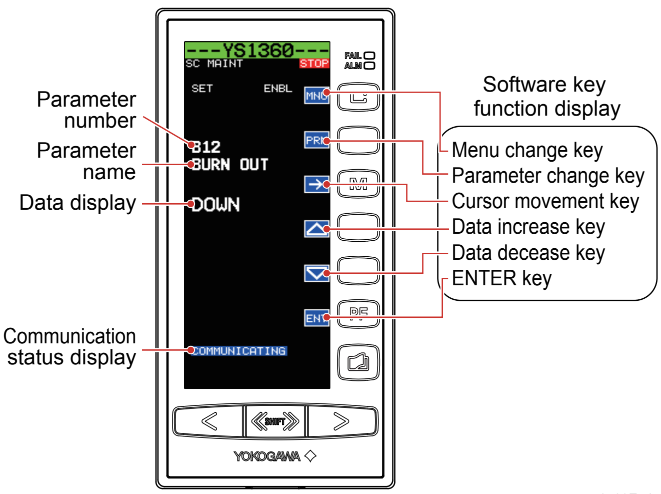
Description

A burnout can be set as for mV input, thermocouple, RTD, and potentiometer input cards.

Setting Details

Operation Display > [SHIFT]+[ ] keys (to the Tuning Menu Display) > [SHIFT]+[ ] keys (to the Engineering Menu Display) > [**SC MAINT**] software key (Input Specification Setting Display)

Engineering Display



Setting Procedure

(example of setting to RTD inputs)

(1)Press the [→] software key to display [SET INHB] in highlight.

(2)Press the [△] software key to display [SET ENBL].

(3) Press the [MNU] software key to display

01

MODEL

Directly input the card model (e.g., ER5*C)

(4)Press the [MNU] software key several times to display

B00

SET

(5)Press the [PRM] software key several times to display the left display.

(6)Press the [△] or [▽] software key to change the data display section.

(7)Press the [ENT] software key to display the data display section in highlight.

(8)Press the [ENT] software key again to accept the setting.

Setting completed.

3.4 Making Zero and Span Adjustments of Input

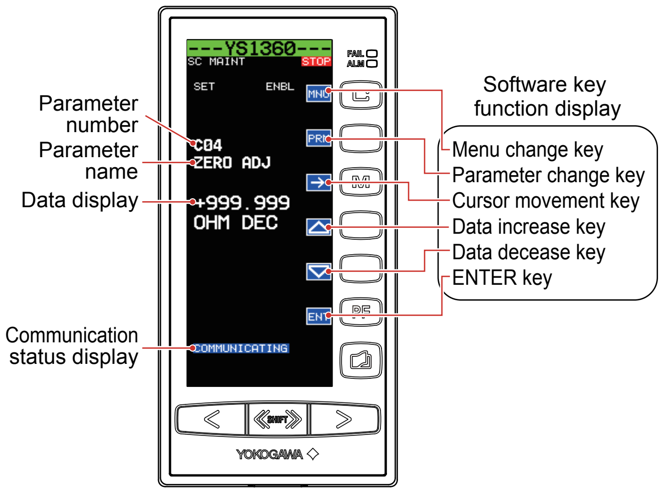
Description

Zero and span adjustments of inputs can be set as for mV input, thermocouple, and RTD cards.

Setting Details

Operation Display > SHIFT + keys (to the Tuning Menu Display) > SHIFT + keys (to the Engineering Menu Display) > [**SC MAINT**] software key (Input Specification Setting Display)

Engineering Display



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Setting Procedure

(1)Press the [→] software key to display [SET INHB] in highlight.
(2)Press the [△] software key to display [SET ENBL].
(3)Press the [MNU] software key to display
01
MODEL
Directly input the card model (e.g., ER5*C)
(4)Press the [MNU] key several times to display
C00
ADJUST
(5)Press the [PRM] software key several times to display the left display.
(6)Press the [△] or [▽] software keys to change the data display section.
(7)Press the [ENT] software key to display the data display section in highlight.
(8)Press the [ENT] software key again to accept the setting.

Setting completed.

4.1 Processing during Power Failures

Description

YS1350/YS1360 enters a power failure status if a momentary power interruption of about 20 ms occurs in the case of 100 V AC supply voltage, or if that of 1 ms or more occurs in the case of 24 V DC. The operation to be taken after power is restored can be set.

Data storage

YS1350/YS1360 data can be stored by making a setting with keystrokes or by making a setting using communication through YSS1000 Setting Software. Data to be stored are the operation mode, manipulated output variables, setpoints, setting parameters, event display data. Trend data displayed on a TREND Display will be lost in the event of a power failure.

Operation after power restoration

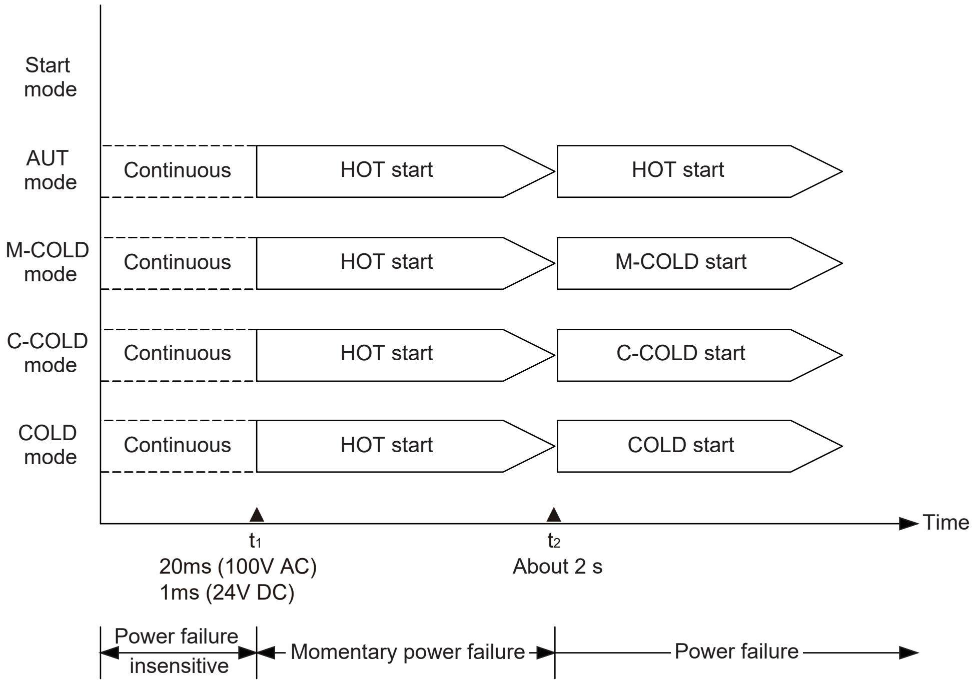
The operation to be taken after power restoration depends on the duration of the power failure and the operation mode that the YS1350/YS1360 is in after power restoration (Start mode (START), engineering parameter).

Start Mode (START)	Power Failure Duration	
	Less than about 2 seconds	About 2 seconds or more
AUT	HOT start	
M-COLD	HOT start	M-COLD start
C-COLD	HOT start	C-COLD start
COLD	HOT start	COLD start

Operation in Each Start Mode

	Start Mode (START)				
	HOT Start	M-COLD Start	A-COLD Start	C-COLD Start	COLD Start
C, M status	Same as before power failure	M mode (LOOP 1)	A mode (LOOP 1)	C mode (LOOP 1)	Same as before power failure
Manipulated output variable (MV) (YS1360)	Same as before power failure	–6.3%			
Setpoint (SV) (YS1350)	Same as before power failure				
Parameters	Same as before power failure				
Process alarms, and system alarms	Continued as is	OFF			
Analog output terminal Y1 (YS1360)	Continued as is	-20%			
Analog output terminal Y2	Continued as is	-6.3%			
DO1 to DO2, DO4	Continued as is	OFF			

Power Failure Duration and Operation in Each Start Mode



[Failure insensitive region (power failure duration < t1)] : t1 = about 1 ms (for 24 V DC type)
: t1 = about 20 ms (for 100 V AC type)
YS1000 performs the same operation as that of continuity status.

[Momentary failure region (t1< power failure duration < t2)] : t2 = about 2 s
The instrument stops operation during power failure.

[Power failure region (t2 < power failure duration)]:
The instrument stops operation during power failure.

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Setting Details

Parameter	Name	Setting Range	Display Transition and Display Title
START	Start mode	AUT: HOT start M-COLD: Power failure duration < 2 sec. ; HOT start, Power failure duration ≥ 2 sec. ; M-COLD start C-COLD: Power failure duration < 2 sec. ; HOT start, Power failure duration ≥ 2 sec. ; C-COLD start COLD: Power failure duration < 2 sec. ; HOT start, Power failure duration ≥ 2 sec. ; COLD start	Tuning Display > Engineering Display > [**CONFIG1**] (Configuration Display1)

WARNING

For products with optional code /FM or /CSA:

(1) Devices must be maintained by professionally trained personnel or ask YOKOGAWA's sales office or sales representative.

(2) In case of option code /FM, install devices according to NEC (National Electrical Code: ANSI/NFPA-70).
In case of option code /CSA, all wiring shall comply with Canadian Electrical Code Part I and local electrical codes.

CAUTION

If the instrument’s front panel becomes soiled or dusty, wipe it gently using a dry, soft cloth. Do not use organic solvents, chemicals, or chemically treated dust cloths. Doing so may result in the instrument case becoming deformed or discolored.

5.1 Inspecting Indication Accuracy

As a guideline, indication accuracy should be inspected on an annual basis.

5.1.1 Calibration Instruments

Name	Description	Number of Units
DC voltage standard	Yokogawa 7651 or equivalent	1
Digital multimeter	Yokogawa 7561 or equivalent	1

5.1.2 Inspecting Input Indication Accuracy

Follow the procedure below to specifically check input indication accuracy.

(1)Apply a voltage of 1.0 V DC to the analog input terminals using the voltage standard.
(2)On the Input and Output Data Display of the Tuning Display, check that the analog input signal concerned is equivalent to 0 ±0.1% in the engineering unit.
(3)Similarly, apply a voltage of 5.0 V DC to check that the analog input signal concerned is equivalent to 100 ±0.1% in the engineering unit.

Parameter	Name	Display Range	Display Transition and Display Title
X1	Analog input 1	−25.0 to 125.0	Tuning Display > **[I/O DATA]** (Input and Output Data Display)
X2	Analog input 2	−25.0 to 125.0	

5.1.3 Inspecting Output Indication Accuracy

Follow the procedure below to specifically check output indication accuracy.

(1)Connect the digital multimeter to the terminals in the current mode if the analog output terminals are current outputs, or in the voltage mode if they are voltage outputs.
(2)Set the operation mode to manual control.
(3)On the LOOP1 Display, manipulate the MV and set the MV1 to 0%.
(4)Verify that the reading is 4 mA DC for current output, or that it is 1.0 V DC for voltage output. (Tolerance is ±0.2% for current output or ±0.1% for voltage output.)
(5)Similarly, set the MV1 to 100% to check that the reading is 20 mA DC for current output and that it is 5.0 V DC for voltage output. (Tolerance is ±0.2% for current output or ±0.1% for voltage output.)

Parameter	Name	Display Range	Display Transition and Display Title
Y1	Analog output 1 (*1)	−20.0 to 106.3	Tuning Display > [**I/O DATA**] (Input and Output Data Display)
Y2	Analog output 2	−6.3 to 106.3	

*1: Displayed on only YS1360.

5.2 Recommended Part Replacement Period

The following shows the replaceable part of the YS1350/YS1360 and the recommended replacement period.

Replaceable Part	Recommended Replacement Period
LCD Display Assembly	About 8 years

WARNING

Part replacement should be carried out by a YOKOGAWA engineer or an engineer certified by YOKOGAWA, as safety standard inspection is required. Contact YOKOGAWA's sales office or sales representative when replacing the parts.

CAUTION

Notes regarding parts with finite life spans

(1) Parts with finite life spans refer to those in which the abrasion or failure period is expected to be reached within 10 years under normal operating or storage conditions. Therefore, parts with life spans of more than 10 years in terms of design are not mentioned here.

(2) The recommended replacement period establishes the period at which preventive maintenance is to be conducted on parts with finite life spans. It does not constitute assurance against incidental failure.

(3) The recommended replacement period is only a guideline and differs depending on operating conditions.

(4) The recommended replacement period may be changed according to field records, etc.

5.3 Packaging when Shipping the Product for Repair

Should the instrument break down and need to be shipped to our sales representative for repair, handle it as noted below:

WARNING

Prior to shipping the instrument, put it into an antistatic bag and repackage it using the original internal packaging materials and packaging container.

6.1 General Specifications

For specifications, see the YS1350 Manual Setter for SV Setting, YS1360 Manual Setter for MV Setting Operation Guide (IM 01B08E02-01EN).

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Revision Information

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